

YOURSTORY



H2S



COCO CLI

HACKATHON

GCC Edition



Team Name: [TRidents](#)

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Team Size: 4

Problem Statement: Predictive Maintenance and OEE Command Center

TRIDENT OPS

EVIDENCE-FIRST PREDICTIVE MAINTENANCE +
OEE

Two evidence lanes. One auditable decision path. Zero fabricated joins.

01 · PROBLEM BRIEF

The gap isn't a model. It's a missing thread.

PIADE

Real line states
alarms + package counts
weighted OEE

NO-JOIN FIREWALL

different plants · no shared key

MetroPT

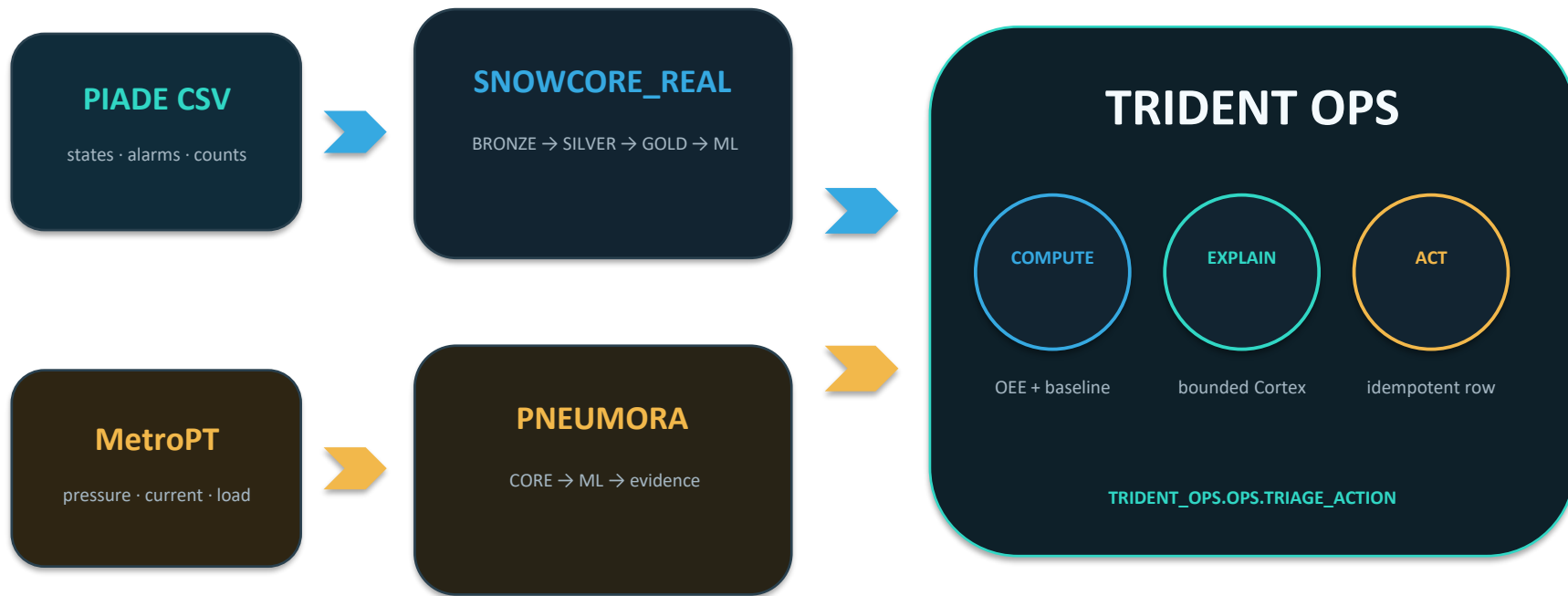
Real compressor sensors
+ failure windows
no production counts

The honest product

PIADE drives OEE, ownership and next-hour ranking. MetroPT remains a separate sensor-evidence lane labelled detection—not forecast.

02 · ARCHITECTURE

One console. Two databases. Three working times.



03 · MEASURED IMPACT

Evidence before ego

46.78%

PIADE weighted OEE

A 64.32% × P 72.93% × Q 99.72%

76.4%

Blind top-decile precision

Persistence baseline 62.4%; archived Dec-2021 holdout

+72 / +86

MetroPT external timing

minutes after leak onset · NO PROMOTION

What changed because of the model?

Not yet measured. OEE lift and realised savings require intervention data.

What did automation prove?

45 replay orders · 35.6% hit rate · 1.26× random timing · 23% coverage

What did search teach us?

25 trials improved validation AUC by 0.0001. Missing signal—not another model—is the ceiling.

The 4-minute proof: compute → explain → act



COMPUTE

CoCo queries weighted OEE and ownership.
Streamlit shows the same deterministic row.



EXPLAIN

Five alarm-pattern rows go to Cortex.
The answer cites codes and states what is unknown.



ACT

The operator opens an inspection.
A Snowflake MERGE writes one reload-safe action.

BLACK-BOX RESULT

same input → same action id → second run deduplicated=true

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THANK YOU

LIVE [TRIDENT OPS](#) CODE github.com/ashokjangam/snowcore-pdm

Compute. Explain. Act. Admit what the evidence cannot prove.